

# Machine Learning (EE526/RA506)

## Mid-Semester Exam

Date: 24/02/2025

Time: Two Hours

Full Marks: 100

[Q1] Evaluate the *Weighted Median* from the following dataset of variates  $x_i$  associated with weights  $w_i$  ( $i = 1, \dots, 10$ ). [5]

|     |   |    |   |    |   |    |    |   |    |   |
|-----|---|----|---|----|---|----|----|---|----|---|
| $x$ | 6 | -4 | 2 | 10 | 0 | 12 | -2 | 4 | -6 | 8 |
| $w$ | 4 | 2  | 3 | 2  | 2 | 4  | 3  | 4 | 3  | 3 |

[Q2] Compute the Mahalanobis Distance between the following two one-dimensional point sets  $S_1$  and  $S_2$ . [5]

|       |    |    |    |    |    |
|-------|----|----|----|----|----|
| $S_1$ | -4 | 4  | -2 | 0  | 2  |
| $S_2$ | 8  | 10 | 9  | 14 | 13 |

[Q3] We are minimizing the objective function  $g(x_1, x_2) = (x_1 - 2)^2 + (x_2 - 3)^2 - 10$  using stochastic search with a solution population of size 5. Consider the following parent solution population at the iteration  $t$  as  $S_t = \left\{ \begin{bmatrix} -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}$ . Report the Evaluated Objective Function Set  $Y_t$ , Solution Fitness Set  $P_t$ , and Number of Children Set  $M_t$ . [20]

[Q4] Let  $n_i$ ,  $\mu_i$  and  $\sigma_i$  be the respective size (number of points), mean and standard deviation of the dataset  $Z_i$  ( $i = 1, 2$ ). The dataset  $Z_i$  is obtained by projecting  $d$ -dimensional points of dataset  $D_i$  along the unit direction vector  $\hat{\omega}$ .

$$D_i = \left\{ \mathbf{x}_k^{(i)} : \mathbf{x}_k^{(i)} \in \mathbb{R}^d; k = 1, \dots, n_i \right\}$$
$$Z_i = \left\{ z_k^{(i)} : z_k^{(i)} = \hat{\omega}^T \mathbf{x}_k^{(i)}; \mathbf{x}_k^{(i)} \in D_i; z_k^{(i)} \in \mathbb{R}^1; k = 1, \dots, n_i \right\}$$

The direction vector  $\hat{\omega}$  is to be optimally chosen for maximum discrimination of the projected pointsets  $Z_1$  and  $Z_2$ . The objective function for this task is as follows.

$$J(\hat{\omega}) = \frac{(n_1 + n_2)(\mu_1 - \mu_2)^2}{n_1\sigma_1^2 + n_2\sigma_2^2}$$



Deduce the expression for  $\hat{\omega}$  that maximizes  $J(\hat{\omega})$ . [15]

[Q5] Consider the following dataset. Suggest a threshold  $T$  for this dataset using Otsu's method. Start iterations with initial threshold  $T_0 = 0$ . [5]

|    |    |    |   |    |    |   |    |    |    |
|----|----|----|---|----|----|---|----|----|----|
| -3 | 31 | 56 | 0 | 28 | -1 | 9 | 24 | -4 | 58 |
|----|----|----|---|----|----|---|----|----|----|

[Q6] Perform K-Means ( $K=3$ ) clustering for the dataset  $X$  and Fill up the following table while estimating Labels  $L$  and Centroids  $M$  for each iteration. Iterate till convergence and report the final labels and the cluster centroids in the following tabular form. [20]

| #              | X  |    |    |    |    |    |    |    |    |   | M <sub>1</sub> | M <sub>2</sub> | M <sub>3</sub> |
|----------------|----|----|----|----|----|----|----|----|----|---|----------------|----------------|----------------|
|                | -5 | 23 | 27 | -2 | 55 | -1 | 57 | 30 | -4 | 8 |                |                |                |
| L <sub>0</sub> | 1  | 2  | 3  | 1  | 2  | 3  | 1  | 2  | 3  | 1 |                |                |                |
| L <sub>1</sub> |    |    |    |    |    |    |    |    |    |   |                |                |                |
|                |    |    |    |    |    |    |    |    |    |   |                |                |                |

[Q7] The dataset  $S = \{x_i; x \in \mathbb{R}^d; i = 1, \dots, n\}$  ( $n = 210000, d = 10$ ) is indexed by using a binary tree. This is realized through a fully-grown binary tree which terminates with at least  $m = 100$  points at each leaf node. This binary tree is constructed by using hierarchical K-Means ( $K = 2$  at each depth). The root node is considered to be at depth  $D = 0$ . (a) What value of maximum depth  $D_{max}$  should be set for constructing this tree. (b) Estimate the total number of parameters required for parametrizing all the split nodes of this fully-grown tree. (c) Consider the task of searching the point  $x \in S$  nearest to the query point  $q \in \mathbb{R}^d$ . Estimate the minimum number of Multiplications, Additions and Comparison operations, if this search is executed by this Two-Means tree learned over  $S$ . [5 × 3 = 15]

[Q8] Consider the problem of incremental clustering. At the  $t^{th}$  instant, the data vector  $x_t$  is used to update the  $k^{th}$  cluster  $G_k^{(t-1)}$  parameterized by cluster mean  $\mu_k^{(t-1)}$ , covariance matrix  $C_k^{(t-1)}$  and the normalized weight  $\pi_k^{(t-1)}$ . Deduce the expression for updating  $\{\mu_k^{(t-1)}, C_k^{(t-1)}, \pi_k^{(t-1)}\}$  to  $\{\mu_k^{(t)}, C_k^{(t)}, \pi_k^{(t)}\}$  with new data  $x_t$ . [3 + 10 + 2 = 15]

----- All the Best -----